

Literature Status: L_1 -Distortion of Wasserstein Spaces over Iterated Laakso Graphs

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Status: literature already answered. Remark 2 of arXiv:2208.13879 records that no nontrivial L_1 -distortion lower bound was known for the 1-Wasserstein spaces over iterated Laakso graphs. Corollary 4.21 of the separate later paper arXiv:2602.14852 proves the sharp linear-in-level lower bound

$$c_1(\mathrm{TC}(\mathrm{La}^{\otimes n})) \geq \frac{2 - \sqrt{2}}{8} n.$$

1 Original open problem

F. Baudier, C. Gartland, and Th. Schlumprecht, *L_1 -distortion of Wasserstein metrics: a tale of two dimensions*, arXiv:2208.13879 [1], prove a general lower-bound theorem requiring isoperimetric and Lipschitz-spectral dimension at least 2. Immediately after that theorem, Remark 2 on arXiv PDF page 3 states:

For graphs whose dimensions are strictly between 1 and 2, such as the Laakso graphs $\mathrm{La}_1^{\otimes n}$, the authors do not know how to prove nontrivial lower bounds for the L_1 -distortion of the associated 1-Wasserstein spaces.

The paper later computes the relevant Laakso dimension as $3/2$, so this case lies genuinely outside the method used there.

2 Separate later answer

C. Gartland and M. Ostrovskii, *Lower Estimates for L_1 -Distortion of Transportation Cost Spaces*, arXiv:2602.14852 [2], explicitly identify the Laakso case as previously unresolved on arXiv PDF page 5, in the paragraph following Theorem 1.2. Their Corollary 4.21 on arXiv PDF page 26 proves, for every $n \geq 0$,

$$c_1(\mathrm{TC}(\mathrm{La}^{\otimes n})) \geq \frac{2 - \sqrt{2}}{8} n.$$

Their notation $\mathrm{TC}(X)$ denotes the transportation-cost, or Lipschitz-free, space over X . Section 5 of the same paper records the standard equality between the L_1 -distortion of the earth-mover (1-Wasserstein) space and the linear L_1 -distortion of $\mathrm{TC}(X)$. Thus Corollary 4.21 directly gives the lower bound requested in the source remark.

3 Scope and status

The later result is stronger than mere unboundedness: since $\log |V(\text{La}^{\otimes n})| = \Theta(n)$, it attains the universal order $\Theta(\log |V|)$. Hence no part of the source’s request for a nontrivial Laakso lower bound remains open at the level stated there.

This identification was found by searching the run’s local full-text arXiv corpus for the exact Laakso/Wasserstein/ L_1 -distortion combination and then checking Theorem 1.2, Corollary 4.21, and the reduction in Section 5 of arXiv:2602.14852. The supporting authors explicitly say the Laakso unboundedness was not known before their work. No new mathematical result is claimed in this packet.

References

- [1] F. Baudier, C. Gartland, and Th. Schlumprecht, *L_1 -distortion of Wasserstein metrics: a tale of two dimensions*, J. Funct. Anal. **284** (2023), 109846; arXiv:2208.13879.
- [2] C. Gartland and M. Ostrovskii, *Lower Estimates for L_1 -Distortion of Transportation Cost Spaces*, arXiv:2602.14852 (2026).